

Product Requirements Document (PRD) AI Multilingual Rural Health Triage Assistant

1. Product Overview

An AI-powered multilingual medical triage assistant designed to help rural users understand symptoms, assess risk, and decide when to seek medical care. The system does not replace doctors but guides users toward appropriate action.

2. Objectives

- Predict possible diseases from symptoms
- Estimate risk level (Low / Moderate / High / Emergency)
- Provide actionable guidance and first aid
- Support multilingual chatbot interface
- Enable rural healthcare accessibility

3. Target Users

Primary: Rural population, low health literacy users, non-English speakers.
Secondary: Community health workers, NGOs, telemedicine platforms.

4. Core Features

- Multilingual chatbot symptom intake
- AI disease prediction (top 3)
- Risk stratification engine
- Emergency detection override
- Action guidance and first aid
- Explainable AI outputs
- Confidence visualization
- Medical disclaimer

5. System Architecture

Frontend: Next.js, React, TypeScript, Tailwind.

Backend: FastAPI, Python, ML model (.pkl).

Infrastructure: Docker, Vercel (frontend), Render/Railway (backend).

ML Layer: Disease classifier with probability outputs.

6. User Flow

1. User selects language
2. Chatbot collects symptoms
3. Backend processes prediction
4. Risk calculated and emergency checked
5. Results displayed with guidance

7. Functional Requirements

Backend: Prediction API, risk engine, emergency detection.

Frontend: Chat UI, results visualization, language switch.

ML: Top 3 predictions with confidence scores.

8. Non-Functional Requirements

- Response time under 2 seconds
- Mobile responsive
- Scalable deployment
- Secure API design

9. Dataset Requirements

- 500 diseases dataset
- Symptom binary features
- Age, gender, duration
- First aid mapping

Target column: Disease

10. Risk & Compliance

Include medical disclaimer: This system is not a medical diagnosis tool. Consult a qualified healthcare professional.

No personal data storage required.

11. MVP Scope

Must Have:

- Chatbot input
- Disease prediction
- Risk levels
- Guidance output
- Emergency detection
- Multilingual UI

12. Milestones

Phase 1: Backend + ML model

Phase 2: Frontend integration

Phase 3: Testing and demo preparation

13. Success Metrics

- Prediction accuracy
- Response speed
- User comprehension
- Demo clarity

14. Elevator Pitch

We built a multilingual AI health triage assistant that helps rural users understand symptoms, assess risk, and decide when to seek medical care — enabling early intervention and potentially saving lives.